

Lab: Explore a Simple Generative Tool

Estimated time needed: 30 minutes

Overview

Generative AI models have revolutionized how you interact with technology, enabling you to create new content, generate realistic images, and translate languages with remarkable accuracy.

In this lab, you will gain hands-on experience with a simple generative AI tool, DataRobot, exploring its capabilities and applications.

Learning Objectives

After completing this lab, you will be able to:

- Sign up in DataRobot
- Add a data set to the use case
- Work on model building

Task 1: Sign-up in DataRobot

Step 1: Click www.datarobot.com

Step 2: Fill in the required information under the "Start your trial today" section and create an account. Once completed the information select "Submit". It will redirect to the signup page.



Try DataRobot for free

Experience how agents, applications, and an end-to-end suite of production-ready tools can accelerate your production use cases

Ship a production-grade agent, app, or PoC in 14 days

Deliver impactful use cases with pre-built AI agents and applications, and explore flexible code-first and GUI workflows:

- Clone and build the app UI, underlying models, and production-ready components with a single line of code
- Create a simple agentic workflow by securely connecting to systems of record and adding additional AI tools
- Customize all pre-built components based on your specific use case and data in DataRobot hosted notebooks or GUI

Quickly deliver AI that impacts your business

■ Faster model delivery

Start your trial today

First Name: *

Last Name: *

Job Title: *

Company: *

Email Address: *



Note: To access the DataRobot platform, you must sign up using a work email address. If you do not have a relevant work email, an alternative is to create a GitHub account using your Gmail address. Once registered, you can log in to DataRobot using your GitHub credentials.

For step-by-step guidance on creating a GitHub account, please refer to the following link:

[GitHub Account Setup Guide](#)

Step 3: A new window will open; select the relevant option for signing in.

Welcome back to DataRobot

 Sign in with Google

 Sign in with GitHub

or

Email

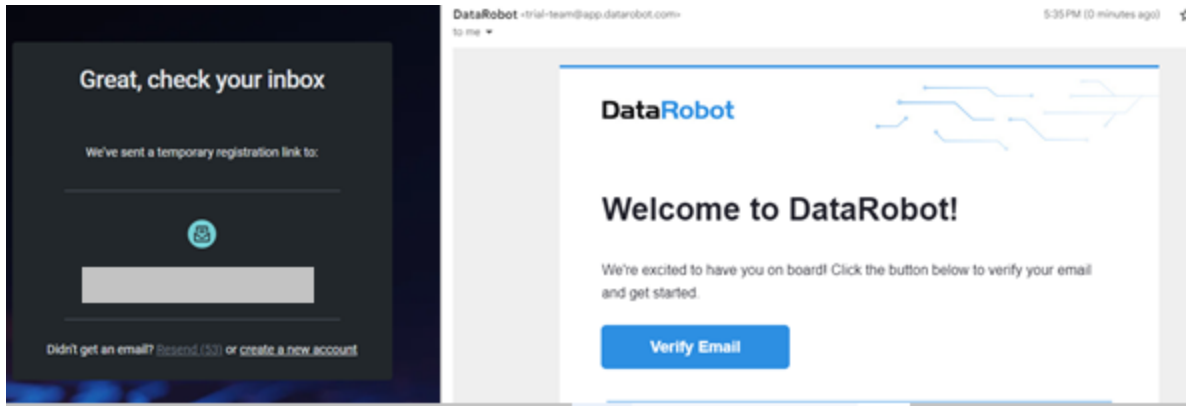
Password

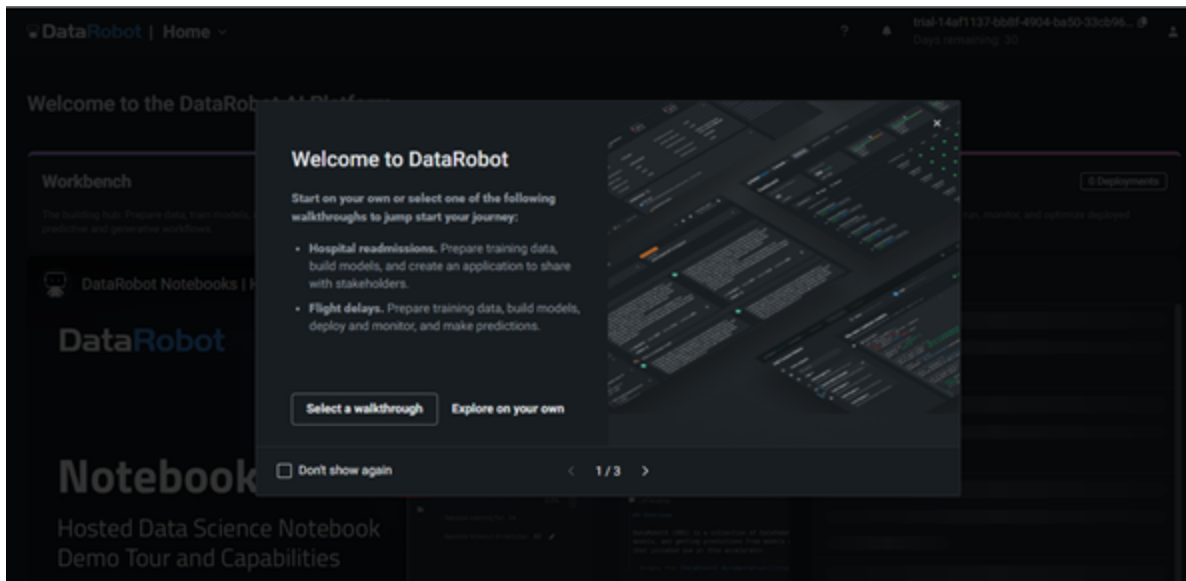
Reset password

Sign in

Step 4: Confirm your email by clicking **Verify Email** in your inbox.



Step 5: Sign up and start your first experience of using the Generative AI tool. The dashboard will look like the image below. You may like to familiarize yourself with the application by clicking **Select a walkthrough**.



Task 2: Add a data set

Step 8: The dashboard will appear shortly, and your screen will look as shown below. Click **Workbench**.

DataRobot | **Workbench** | Registry | Console

Shortcuts to AI success
Explore our recommended actions to kickstart your AI journey and unlock the full potential of your data.

Explore application templates
Accelerate development with end-to-end solutions. These code-first, reusable pipelines are available out-of-the-box, but also offer easy customization, for quick, tailored successes.
[Explore application templates](#)

Build ML and GenAI models
From data to model training to insights, build experiment-based, iterative workflows for predictive and generative AI models.
[Explore Use Cases](#)

Recent activity

Workbench | Registry | Console

Name	Use Case	Type	Last modified	Last i
------	----------	------	---------------	--------

Get started
More walkth
[GenAI walkth](#)
Create and cor
Augmented Ge

Step 9: Click **Create Use Case**.

DataRobot | Workbench

trial-14af1137-bb8f-4904-ba50-33cb96...
Days remaining: 30

Use Case directory

[+ Create Use Case](#)

LAST MODIFIED USE CASES

Untitled_3

P P
Created just now

0 0 0 0 0 0

Hospital Readmission

Walkthrough

P P
Modified 1 min ago

2 0 0 0 0 0

Flight Delays

Walkthrough

P P
Modified 5 min ago

1 0 0 0 0 0

ALL USE CASES 3

Search

Name	Datasets	Vector databases...	Experiments	Playgrounds	Apps	Notebooks	Created by	
F_ Walkthrough	1	0	0	0	0	0	P P	I
H_ Walkthrough	2	0	0	0	0	0	P P	I

Step 10: Click **Data assets** and select **Browse Data** to include the data set in your use case.

DataRobot

Workbench

Registry

Console

trial-bbec95-ba16-46b1-aded-11f...

Days remaining: 30

←

Use Case directory / **Untitled_3** ▾

Use Case management

Use Case assets

Data assets

Vector databases

Experiments

Playgrounds

Notebooks & codespaces

Applications

More

Data

Data associated with this Use Case

There's nothing here yet. Drag-and-drop data here or select an option below to import data for use in experiments and vector databases.

Browse data

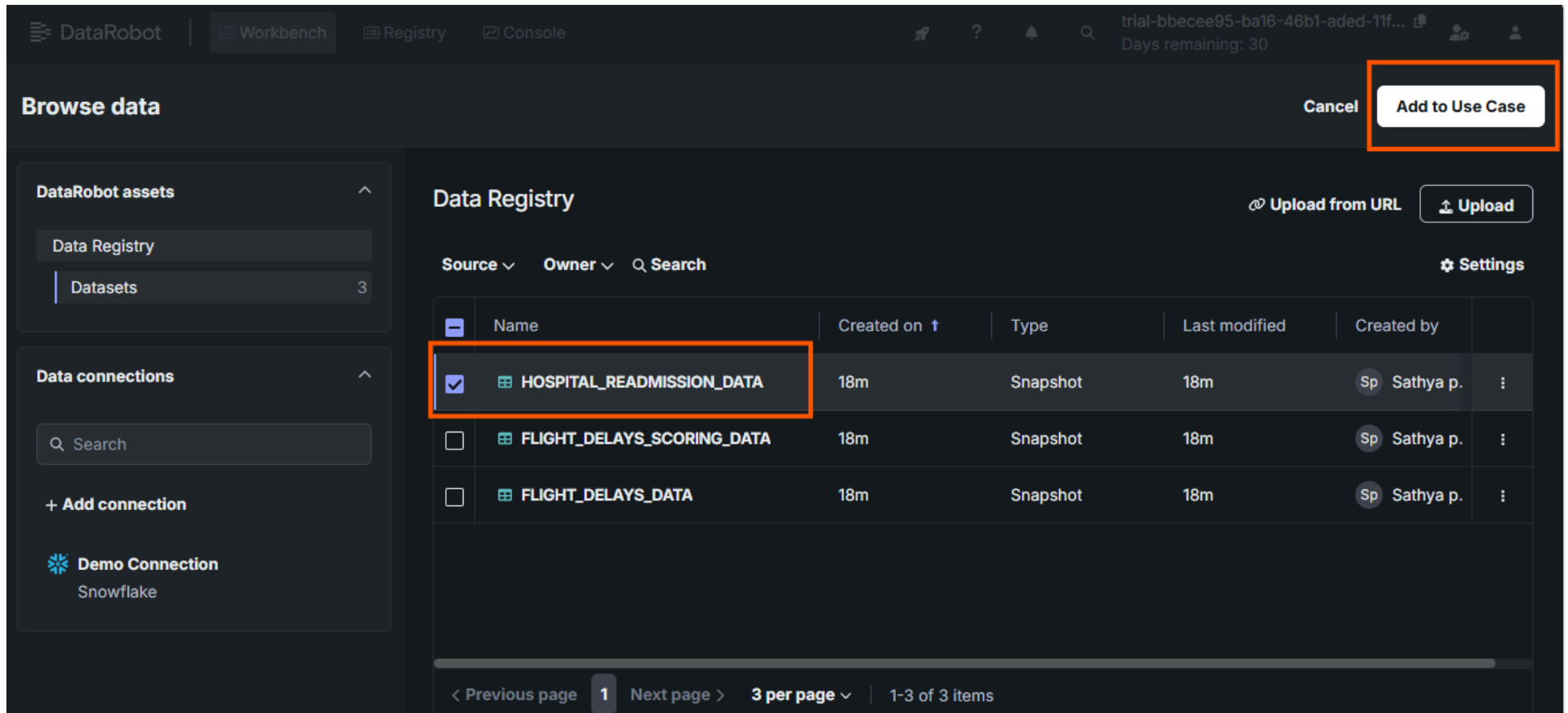
Upload file

Add from URL

Accepted formats: .csv, .tsv, .dsv, .xls, .xlsx, .sas7bdat, .geojson, .gz, .bz2, .tar, .tgz, .zip, .parquet

[View dataset requirements](#)

Step 10: You can select an in-built sample data set *HOSPITAL_READMISSION_DATA* and select **Add to Use Case**.



The screenshot shows the DataRobot interface. On the left, there's a sidebar with 'DataRobot assets' and 'Data connections'. The 'Data Registry' is selected under 'DataRobot assets'. The 'Data connections' section shows a 'Demo Connection' for 'Snowflake'. The main area displays the 'Data Registry' with a table of datasets. The 'HOSPITAL_READMISSION_DATA' dataset is selected and highlighted with an orange box. In the top right corner, the 'Add to Use Case' button is also highlighted with an orange box.

Browse data Cancel Add to Use Case

DataRobot assets

- Data Registry
- Datasets 3

Data connections

Search

+ Add connection

 **Demo Connection**
Snowflake

Data Registry Upload from URL Upload Settings

Source Owner Search

	Name	Created on ↑	Type	Last modified	Created by	
☒	 HOSPITAL_READMISSION_DATA	18m	Snapshot	18m	Sp Sathya p.	:
☐	 FLIGHT_DELAYS_SCORING_DATA	18m	Snapshot	18m	Sp Sathya p.	:
☐	 FLIGHT_DELAYS_DATA	18m	Snapshot	18m	Sp Sathya p.	:

< Previous page 1 Next page > 3 per page 1-3 of 3 items

Step 11: Once you select the data set, you can see a preview of it by clicking on the dataset. You can also view the data set's features, as shown below. Click **Add to Use Case**.

DataRobot | Workbench Workbench directory / Untitled_4 trial-c3398e81-5e83-4e52-b732-aca1c... Days remaining: 29

Add data Close Add to Use Case

DATA STORES

↑ Upload + Connect

Search

Data Registry
DataRobot

Demo Connection
Snowflake

HOSPITAL_READMISSION_DATA
Data Registry / HOSPITAL_READMISSION_...

Features Data preview

race	gender	age	weight	admission_type_id	discharge_dispositi
Categorical	Categorical	Categorical	Categorical	Categorical	Categorical
Caucasian	Male	[50-60)		Emergency	Discharged to hom
Caucasian	Female	[70-80)		Urgent	Discharged/transfe
Caucasian	Male	[60-70)		Elective	
Caucasian	Male	[60-70)			Discharged to hom
Caucasian	Female	[40-50)		Not Available	Discharged to hom
Caucasian	Male	[90-100)		Emergency	Discharged/transfe
Caucasian	Male	[70-80)		Urgent	Discharged/transfe
Caucasian	Female	[80-90)		Urgent	Discharged to hom
Hispanic	Female	[50-60)		Emergency	Discharged to hom

Snapshot sample 51 features | 1,511 rows

Step 12: After you add the data set to the use case, the workbench will appear as shown below. You can click the data set to see the feature insights.

DataRobot | **Workbench** ▾ Workbench directory / **Untitled_4** ▾ ? 🔔 trial-c3398e81-5e83-4e52-b732-aca1c... 👤
Days remaining: 29

< Use Case directory

USE CASE
Untitled_4 ⋮ **Add Data** ▾

All **Data** ① **Vector databases** **Experiments** **Playgrounds** **Applications**

Type ▾ 🔍 Search ⚙️ Settings

Name	Created By	Last Modified ↑	Type	Source	Rows	
 HOSPITAL_READMISSION_DATA	 Pratiksha V.	now	Snapshot	Snowflake	10000	⋮

< Previous **1** Next > **1 per page** ▾ 1-1 of 1 items

Use Case info

DESCRIPTION
Add a description

CREATED
July 29, 2024

LAST MODIFIED
just now

OWNER
 **Pratiksha Verma**

TEAM
None

Manage members

YOUR ROLE
Owner

Step 13: Explore the **All Features** menu to display specific features.

DataRobot | Workbench Workbench directory / Untitled_4 / **HOSPITAL_READMISSION_DATA**

Jul 29th, 2024 10:39 AM Snapshot Data actions

HOSPITAL_READMISSION_DATA

Data preview **Features** Feature lists

Show insights Show features from: All Features + Create feature list

Search

DATAROBOT FEATURE LISTS

Feature List	Count
All Features	51
Informative Features	40
Raw Features	51

race
Categorical

age
Categorical

weight
Categorical

admission_type_id
Categorical

discharge_status
Categorical

54% [70-80] 26% ==Missing== 96% Emergency 49% Discharge
21% Male 46% [60-70] 22% [75-100] 2% Urgent 19% Discharge
6% Other 52% Other 2% Other 32% Other

Caucasian
AfricanAmerican
Other

Caucasian
Caucasian
Caucasian

Male
Female
Male

[50-60]
[70-80]
[60-70]

Emergency
Urgent
Emergency

Discharge
Discharge
Discharge

Snapshot sample 51 features | 1,511 rows

Information

DATE ADDED
July 29, 2024

ADDED BY
Pratiksha Verma

FEATURES
51

ROWS
10,000

SIZE
5.48 MB

DATABASE
Snowflake / Demo Connection

SCHEMA
TRIAL_READONLY

TABLE
HOSPITAL_READMISSION_DATA

Dataset Versions

Task 3: Work on Data Modeling

Step 14: You will have options **Modelling** and **Start wrangling**. You can try data wrangling if you want to. For this lab, you will work on model building under **Data actions** and select **Modelling**. It will take a while to prepare a data set for modelling.

DataRobot | Workbench Workbench directory / Untitled_4 / HOSPITAL_READMISSION_DATA

Jul 29th, 2024 10:39 AM Snapshot Data actions ^

Data preview Features Feature lists

☒ Show insights Show features from: All Features + Create feature list

race	gender	age	weight	admission	discharge
Categorical	Categorical	Categorical	Categorical	Categorical	Categorical
Caucasian	Female	[70-80]	==Missing==	Emergency	Discharge
AfricanAmerican	Male	[60-70]	[75-100]	Urgent	Discharge
Other		Other	Other	Other	Other
Caucasian	Male	[50-60]		Emergency	Discharge
Caucasian	Female	[70-80]		Urgent	Discharge
Caucasian	Male	[60-70]		Emergency	Discharge

Snapshot sample 51 features | 1,511 rows

Information

DATE ADDED
July 29, 2024

ADDED BY
Pratiksha Verma

FEATURES
51

ROWS
10,000

SIZE
5.48 MB

DATABASE
Snowflake / Demo Connection

SCHEMA
TRIAL_READONLY

TABLE
HOSPITAL_READMISSION_DATA

Dataset Versions

Step 15: Once done, you need to select the **Target feature**. Select **readmitted** as your target feature.

trial-c3398e81-5e83-4e52-b732-aca1c...

Days remaining: 29

?

DataRobot | Workbench

Workbench directory / Untitled_4

Set up new experiment

Dataset

Target

Additional settings

Exit

< Back

Next >

Target feature

Select the feature to make predictions on.

payer_code

pioglitazone

race

readmitted

reoacolinide

troglitazone

tolbutamide

tolazamide

time_in_hospital

			Uniq...	Missi...	Mean	Std Dev
			7	9592	-	-
troglitazone	35	Categorical	1	0	-	-
tolbutamide	30	Categorical	2	0	-	-
tolazamide	36	Categorical	2	0	-	-
time_in_hospital	7	Numeric	14	0	4.43	3.021

Experiment summary

HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57

Dataset

Name

HOSPITAL_READMISSION_DATA

Rows

10,000

Features

51

Target

No target selected

Step 16: The workbench screen will be displayed as shown below. Click **Next**.

DataRobot | Workbench Workbench directory / Untitled_4

trial-c3398e81-5e83-4e52-b732-aca1c...
Days remaining: 29

Set up new experiment

Dataset Target Additional settings

Exit < Back Next >

Target feature
Select the feature to make predictions on.

readmitted

Target type: Binary classification ⓘ

Positive class: ☐ 0 ☒ 1 ⓘ

Modeling mode
Set the mode used for selecting which blueprints to build when training models.

Quick Autopilot

Optimization metric
Set the metric used when training models to evaluate and optimize accuracy.

LogLoss (Accuracy) Recommended

Value	Number of rows
False	~5800
True	~3800

Experiment summary

HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57

Dataset

Name	HOSPITAL_READMISSION_DATA
Rows	10,000
Features	51

Target

Feature	readmitted
Target type	Binary classification
Positive class	1
Modeling mode	Quick Autopilot
Optimization metric	LogLoss
Training feature list	Informative Features

Partitioning

Step 17: You can modify the model setting in **Additional Settings**; once done, click **Next** and then click **Start modelling**.

trial-c3398e81-5e83-4e52-b732-aca1c...

Days remaining: 29

?

🔔

👤

DataRobot

Workbench

Workbench directory / Untitled_4

Set up new experiment

✓ Dataset

✓ Target

⚙️ Additional settings

Exit

< Back

Start modeling

Data partitioning

Time series modeling

Preview

⚙️ Additional settings

Partitioning method

Select the method for assigning rows to partitions when training models.

Stratified sampling

Rows are assigned to ensure similar target distribution across each partition.

Validation type

☒ Cross-validation

Trains models on a specified number of folds, maximizing data use but also increasing run time.

☐ Training-validation-holdout

Splits data into three partitions: trains models on the training set, assess performance on the validation set, and evaluates the model on unseen data in the holdout set.

Cross-validation folds

Enter a value from 2 - 50.

Holdout percentage

Set the subset of data that is unavailable during training and validation. Enter a value

Experiment summary

HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57

Dataset

Name	HOSPITAL_READMISSION_DATA
Rows	10,000
Features	51

Target

Feature	readmitted
Target type	Binary classification
Positive class	1
Modeling mode	Quick Autopilot
Optimization metric	LogLoss
Training feature list	Informative Features

Partitioning

Step 18: Building models will take a while.

The screenshot displays the DataRobot Workbench interface. At the top, the header includes the DataRobot logo, the 'Workbench' tab, and the current project path: 'Workbench directory / Untitled_4 / HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57'. A user profile icon and a notification bell are also present. Below the header, the left sidebar contains navigation options: 'Experiment' (selected) and 'Comparison'. Under 'Experiment', there are links for 'View experiment info', a 'Filter' button, and a 'Validation' dropdown set to 'LogLoss'. A 'Search' bar is also visible. The 'MODELS' section shows '0/0' models. Two models are listed as 'Building...': 'Keras Slim Residual Neural Network Classifier using...' and 'Elastic-Net Classifier (L2 / Binomial Deviance)'. Both models show 'Informative Features' and '64% (6,400 rows)' of data. The main workspace area is dark and features a large circular progress indicator and the text 'Models are building...'. On the right side, a vertical panel shows system resources: 'CPU' usage, a '+2-' button, and a '2' button.

Step 19: once the modelling is complete, you can pick a model of your choice, and the DataRobot will show the **Model Overview**.

The screenshot displays the DataRobot Workbench interface. The top navigation bar shows the project name 'HOSPITAL_READMISSION_DATA' and the date '2024-07-30 11:23:57'. The left sidebar contains a 'Models' list with three entries: 'Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)' (score 0.6089), 'Elastic-Net Classifier (L2 / Binomial Deviance)' (score 0.6115), and 'Keras Slim Residual Neural' (score 0.6327). The main panel is titled 'Model Overview' and shows details for the selected Elastic-Net Classifier. It includes a table of training scores (Validation: 0.6089, Cross-validation: 0.6111, Holdout: 0.6111) and training settings (Training feature list, Informative Features: 64% (6,400 rows), Training sample size: 64% (6,400 rows)). Below these are expandable sections for 'Blueprint', 'Feature Impact', 'Feature Effects', 'Individual Prediction Explanations', and 'ROC Curve'.

DataRobot | Workbench Workbench directory / Untitled_4 / HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57

Model Overview

Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance) ☆

Training scores: LogLoss

Validation	0.6089
Cross-validation	Score
Holdout	0.6111

Training settings

Training feature list	Informative Features 17
Training sample size	64% (6,400 rows) 17

Models (3 / 3)

Models are building...

- Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)** ☆ 0.6089
 - Informative Features
 - 64% (6,400 rows)
- Elastic-Net Classifier (L2 / Binomial Deviance)** ☆ 0.6115
 - Informative Features
 - 64% (6,400 rows)
- Keras Slim Residual Neural** ☆ 0.6327

Blueprint ⓘ

Feature Impact ⓘ

Feature Effects ⓘ

Individual Prediction Explanations ⓘ

ROC Curve ⓘ

Step 20: You can explore various model overview components like **Blueprint**, **Feature Impact**, and so on.

Experiment

Comparison

View experiment...

Filter 0

Validation • LogLoss

Search

MODELS 3/3

Models are building...

Generalized Additive2...
Building...

Informative Features

64% (6,400 rows)

Light Gradient...
Building...

Informative Features

Blueprint

Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)

One-Hot Encoding | Missing Values Imputed | Standardize | Matrix of word-grams occurrences | Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)

ElasticNet Generalized Linear Model

Click a blueprint node to view more information. View more blueprints in the [blueprint repository](#).

Search

Edit blueprint

Data

Categorical Variables

Numeric Variables

Text Variables

One-Hot Encoding

Missing Values Imputed

Matrix of word-grams occurrences

Standardize

Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)

Prediction

Feature Impact

Data slice: All data

Compute method: SHAP

Sort by: Feature impact • Decr...

Use quick-compute

Search

Actions

0%

50%

100%

number_inpatient

number_diagnoses

medical_specialty

admission_type_id

discharge_disposition_id

admission_source_id

diag_3

diag_1

age

diag_2

num_lab_procedures

number_emergency

diabetesMed

race

insulin

payer_code

change

diag_2_desc

plglltzone

Step 21: If you have test or unseen data, you can also make predictions by clicking **Make Predictions** under **Model actions**.

The screenshot displays the Skills Network Editor interface, divided into two main sections: **Model Overview** and **Make Predictions**.

Model Overview Section:

- Experiment/Comparison:** Tabs at the top left.
- View experim...:** A button with a refresh icon and a trash icon.
- Filter:** A dropdown menu showing "Validation • LogLoss".
- Search:** A search bar.
- MODELS:** A list showing 8/8 models.
- Model actions:** A dropdown menu with options: Register model, Make predictions (highlighted with an orange box), Create app, Generate compliance report, and Delete model.
- Training scores: LogLoss:** A table showing Validation (0.6089), Cross-validation (Score), and Holdout (0.6111).
- Training settings:** A table showing Training feature list, Informative Features (64% (6,400 rows)), and Training sample size.
- Blueprint:** A section with a plus icon.

Make Predictions Section:

- Model Overview:** A back arrow and the text "Model Overview".
- Make Predictions:** The main heading.
- Make new predictions:** A section with a "Prediction dataset" label and a note "Prediction datasets cannot exceed 5.00 GB.".
- Prediction options:** A section with a "Drop file(s) here to upload or Choose file" button, an "Upload local file" button, a "Use model training data" button, and a "Data registry" button.
- Include additional feature values in prediction:** A toggle switch that is currently turned on. Below it, a note states: "When enabled, results include specified feature values taken from the prediction dataset.".
- Add all features / Add specified features:** Two radio buttons. "Add all features" is selected.
- All features selected:** A text box showing the selected features.
- Include Prediction Explanations:** A toggle switch that is currently turned off.
- Compute and download predictions:** A button at the bottom.
- Download recent predictions:** A section with the text "Predictions are stored and available to download for 48 hours." and "No recent predictions".

Step 22: You can also click **Generate compliance report** and **download compliance report** for your use case.

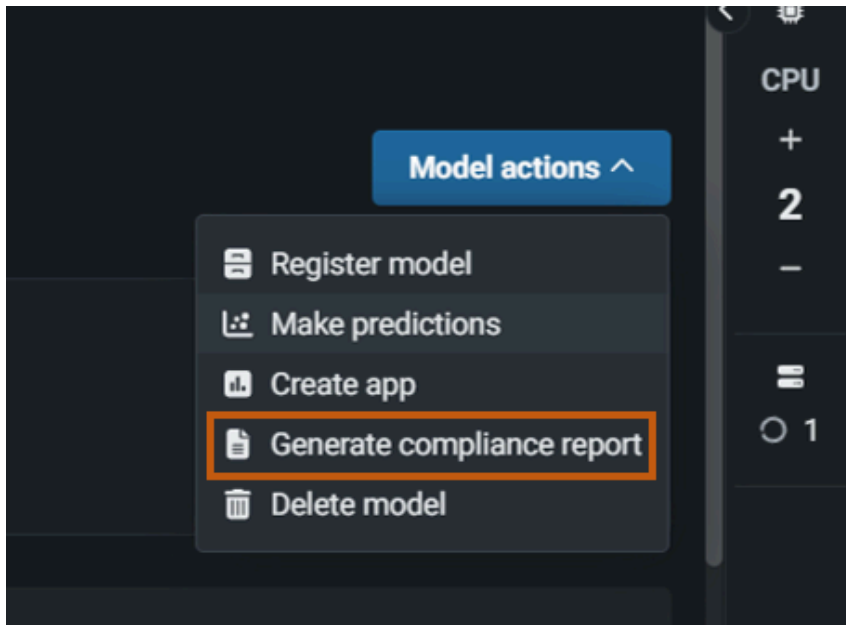


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Conclusion

In this lab, you have signed up in DataRobot, added a data set in a use case, and worked on data modelling.